



Illuminating
ENGINEERING SOCIETY

APPROVED METHOD:
IES SPATIAL DAYLIGHT AUTONOMY
(sDA) AND ANNUAL SUNLIGHT
EXPOSURE (ASE)

AN AMERICAN NATIONAL STANDARD



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Publication of this document
has been approved by IES.
Suggestions for revisions
should be directed to IES.

**Prepared by
The IES Daylight Metrics Committee**



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Preface

Assessing the dynamic qualities of a daylit space requires different methods of assessment than those that have been developed for a space that is electrically illuminated. With electric lighting, average illuminance is a significant and useful metric, especially in designs that aim to provide general illumination at a predetermined target illuminance. However, in a daylit space, average illuminance has less meaning and utility in practice. One reason is purely spatial; for example, sidelit environments inherently have non-uniform illuminance distributions due to the geometric relationship of room and aperture, as well as internal shading from furnishings. Typically, there are high illumination levels near a window, which quickly diminish with increased distance from these daylight apertures.

Other reasons are both spatial and temporal; the daylight sources of sun, sky, and clouds vary in luminous intensity and position each moment of the day and over the course of a year, and corresponding illumination within a space varies relative to the geometry of daylight apertures such as windows and skylights. In addition, it is well known that building occupants can have a profound impact on daylight availability through their operation of window management devices, such as blinds, shades, or variable transmission glass. These may be integral to an original design or, perhaps more frequently, are subsequently retrofitted onto existing windows to improve occupant comfort. Finally, interior daylight illumination can easily be an order of magnitude brighter than the illumination provided by electric lighting, thus shifting the adaptation levels of the eye. Therefore, human perception of illuminance sufficiency in spaces with daylight and/or direct sunlight is more complex than in spaces illuminated only by electric lighting.

Because daylight illumination levels are dynamic, the performance of daylight needs to be considered over time. Evaluation of annual daylight performance integrates these variations over one full year, including both daily and seasonal variations. Because the availability of daylight is highly dependent upon local climate conditions, especially the daily and seasonal

balance of daylight provided from direct sunlight versus the sky and clouds, accounting for local climate conditions is also critical. The optimal design of a daylit system is likely to be very different for a foggy coastline location compared to an inland desert or mountaintop.

Over the last few decades, a variety of daylighting metrics have been proposed and implemented in simulation tools to overcome the inability of older metrics and earlier tools to assess these dynamic conditions. Most of these metrics require substantial computational power to process a large number of input variables, such as the geometry of building, site, and surround; climate data; occupancy schedules; and criteria for operating window management devices to control excessive sunlight. Because the accuracy, variability, and formatting of inputs and outputs can be quite challenging, interpretation of these metrics has become problematic. This document describes a standardized analysis methodology that results in a pair of daylight performance metrics that can be used to compare across designs, building types, and locations.

1.0 Introduction, Changes from Previous, Scope, Overview

1.1 Introduction

This document describes two annual daylight performance metrics, spatial Daylight Autonomy (sDA) and Annual Sunlight Exposure (ASE), which provide two useful dimensions for evaluating daylight performance. Both metrics are generated via a similar computer-based simulation methodology that uses a full year of hourly weather data to calculate illuminance values inside a given architectural space. The sDA metric is distinguished from many others in that it explicitly accounts for the movement of operable shading devices at daylight apertures, which hereafter in this document will be collectively referred to as blinds.

Spatial Daylight Autonomy reports on the level and prevalence of daylight in a space over the course of a year, after the operation of any daylight management devices at the aperture. Thus, sDA addresses the

questions “Will this be a sufficiently daylit space?” and “How much of this space, or building, is illuminated by daylight?”

Annual Sunlight Exposure reports on the amount of direct beam sunlight that could potentially enter a space given its fixed architectural geometry, before the operation of blinds. Thus, ASE addresses the questions “What is the risk of visual discomfort from too much sunlight in this space?” and “Should design changes and additional daylight management devices be considered to reduce this risk?”

1.2 Changes From IES LM-83-12

ANSI/IES LM-83-22 is an update of the previous publication, *IES LM-83-12, Approved Method: IES Spatial Daylight Autonomy (sDA) and Annual Sunlight Exposure (ASE)*.¹ It includes a few modifications and additional details on how these metrics are to be computed, as well as recommended tables for the reporting of simulations to produce these metrics. A brief summary of these is provided in this section, with specific details on these topics included in later sections of this document.

Changes related to sDA involve both changes to the simulation process and to the model requirements. First, a lower illuminance threshold value of 150 lux (15 fc) may now be applied, in place of the standard value of 300 lux (30 fc),* in spaces with less critical visual tasks. In addition, areas or spaces where no critical visual task is performed, or where direct sunlight may be desired (such as areas of a building that are planned to be sunlight tolerant), may be excluded from the daylighting analysis area, but if so, those designated areas shall be excluded from both sDA and ASE reporting. Dirt depreciation factors are now required on all glazing surfaces for simulation of these metrics. This document also clarifies that in simulating sDA, blinds are required to be included on all image-preserving glazing, including skylights. For window groups and the control of blinds in an sDA analysis, windows that are within ± 10 degrees in azimuth

orientation may be included in the same window group, but finer angular separation is permitted. In addition, all glazing area covered by a single blind is required to be placed into the same window group, but any windows that are independently controlled may be assigned to separate window groups. To simplify sDA calculations for daylight apertures with dynamic glazing, an optional simplified approach is provided, which may be applied in the simulations instead of modeling the actual control algorithm that will be applied in the building.

Second, as described in **Section 3.0**, ASE is now reported two different ways to account for the use of automated daylight management systems. The gross ASE considers all image-preserving glazing, while the net ASE considers only the image-preserving glazing areas that do not apply an automated daylight management system (i.e., apertures with, for example, automated blind systems or dynamic glazing are not considered in the calculation of net ASE). If there are no automated daylight management systems at the daylight apertures in a building, then the gross and net ASE values are the same.

Finally, guidelines for the tabular reporting of space and building simulations of the metrics discussed in this document are presented, and a procedure to compute the total sDA of a whole building is provided. These formatted tables may be helpful for performance assessment by the design team as well as for submission for code compliance review or other documentation of daylighting performance, such as required by a rating organization.

1.3 Background and Scope

Spatial Daylight Autonomy and Annual Sunlight Exposure are intended for application to common workplace environments and can be used for evaluating a single space or a collection of spaces across a building. The foundational research² was conducted in open offices, classrooms, meeting rooms, multipurpose rooms, library spaces, and service areas in lobbies. Therefore, the metrics are most applicable to spaces with similar visual tasks and occupancy patterns. The studied spaces were located in the continental United States, and thus are most applicable to similar latitudes (35 to 55 degrees) and for occupants with similar cultural backgrounds and experiences with daylight.

* In 2019-20 the Illuminating Engineering Society implemented 10:1 instead of 10.76:1 lux-footcandle conversion method for the Recommended Practice Illuminance Tables. For IES LM-83, sDA and ASE values will vary depending upon the conversion method used. Lighting practitioners are encouraged to use lux when feasible.

These metrics are calculated for a discrete “analysis area,” as defined in **Section 4.1**, and address performance within a physically defined interior space. The metrics can be used to analyze daylighting performance in an existing building or in a new design, from the early conceptual stage through construction documents. They can also be used to guide design specifications for owners, building codes, and voluntary certification programs, and to inform field research and the design of optimization studies. It is for this reason that a standardized methodology and set of input parameters are presented in this document for use in applying these metrics.

It is important to acknowledge that these metrics are intended to describe an asset value (an assessment applying standardized ratings) for a given building design, based on standardized assumptions, rather than to predict the actual performance that would be determined by real-time weather and concurrent human behavior. “Asset” value refers to the value of a physical building independent of its current occupants’ desires, behavior, and management. It distinguishes between the tangible aspects of building design and the impact of operational variables. In LM-83 many operational defaults are used to define a “normative” condition for general rating purposes, similar to a kilometer per liter (mjiles per gallon) rating for a car.

Producing an acceptable sDA and ASE does not guarantee a “good design,” as these metrics provide only a first order-of-magnitude comparison between spaces and design strategies, and do not address important details related to the space, occupancy periods, visual tasks, building site, or thermal considerations that might affect the quality of a particular design. A more complete design study may require additional analyses or alternate assumptions than are applied in the standardized sDA and ASE metrics. The primary goal of this document, and the metrics it presents, is to create a consistent methodology that facilitates comparable reporting across research, design, and policy endeavors.

There are many things that these daylight performance metrics are not. They, and the illuminance threshold values that apply to them, are not counter to, nor do they supersede, IES task lighting criteria for these space

types found in IES Recommended Practice documents. In addition, these metrics do not directly address human health outcomes, an ongoing area of research. These metrics also do not address energy performance, which would require a detailed analysis that also considered the design and operation of the electric lighting system, occupancy patterns, and the effect of the fenestration system on building heating, cooling, and ventilation. Finally, these metrics do not directly assess disability glare, discomfort glare, or veiling glare, which require a different type of analysis that is based on the luminances in an occupant’s field of view.

1.4 Overview

The list that follows provides an overview of the process that is outlined in this document for the analysis of sDA and ASE across a single space, and across an entire building.

1. When conducting an analysis for a building, the first step is to subdivide the building into individual spaces (such as rooms), which are defined by volumes significantly surrounded by solid surfaces, but may also be separated by lighting design target illuminances, as noted in **Section 4.2.1**, or by facade orientation in the case of large open-plan designs.
2. The second step is to define the non-critical task areas that are to be excluded from an analysis (as described in **Section 4.1.1**). The total area of these spaces is computed for reporting purposes.
3. Next, sunlight tolerant areas are defined. These are areas where uncontrolled sunlight is welcomed. They may be a portion of a space or an entire space, as described in **Section 4.1.1**. The total area of these spaces is also computed for reporting purposes.
4. The remaining building area is the critical task area over which daylight simulations of sDA and ASE will be performed on a space-by-space basis. Analysis grids are located across these spaces, and simulations of ASE and sDA calculations are performed across the analysis points.
 - a. ASE_{gross} is first computed and recorded across all image-preserving glazing. Then the contribution from image-preserving glazing areas that have automated blinds or dynamic glazing for glare control is removed from the gross ASE value to evaluate the ASE_{net} .

- b. The sDA value is computed for three different conditions: with all blinds open, all blinds closed, and with blinds or dynamic glazing operating according to the requirements of **Section 2.2**.

When analyzing an entire building, the total building ASE and sDA are compiled across all critical task areas, first by orientation, then across the entire building.

2.0 Spatial Daylight Autonomy (sDA)

Spatial Daylight Autonomy is a metric that describes the sufficiency of ambient daylight levels in interior environments across the year, while also accounting for the operation of daylight management devices that limit direct sunlight penetration into the space. It describes the fractional *area* of a space that achieves a specified Daylight Autonomy (DA). Daylight Autonomy is the fraction of *time* in an annual simulation or study period for which a given analysis point meets or exceeds a target illuminance level with daylight alone.

Spatial Daylight Autonomy is calculated using a computer simulation that models daylight availability in a given space (i.e., across an analysis area), based on a typical meteorological year (TMY) hourly weather data file from a location as near as possible to the building being studied. These weather files account for solar position, solar intensity, cloudiness patterns, and sky brightness. This is generally referred to as climate-based daylight modeling (CBDM). Daylight Autonomy with a 300 lux (30 fc) illuminance threshold (DA₃₀₀, on which sDA₃₀₀ is based), has shown good agreement with occupants' assessment of the daylighted area in typical spaces.³

Blinds, for the purpose of calculating sDA, are operated to limit the amount of direct sunlight that can enter the space. (As mentioned earlier in this document, the term "blinds" is used as a shorthand for all possible operable daylight management devices that can be applied across a daylight aperture.) Thus, sDA is a conservative value that assumes that the daylight illumination levels in the space are not unduly raised by large areas of

sunlight that might cause visual discomfort. For the purposes of sDA calculations, blinds are closed by window group to limit direct sunlight to 2% or less of the analysis points, as described in **Section 4.6**.

Spatial Daylight Autonomy is defined as the percentage of an *analysis area* that meets a specified *daylight illuminance threshold value* for a specified *time fraction* of the specified *annual analysis* period. The calculation of sDA requires hourly daylight illuminance to be determined at each analysis point across the year, with blinds operating according to a specified algorithm.

2.1 Standardized Inputs, Terms, and Concepts

For sDA values to be understood relative to one another, a means of properly specifying an *analysis area* and a standardized set of simulation parameters is required. For this reason, the *daylight illuminance threshold value*, the *annual analysis hours*, the *time fraction*, and *blinds operation algorithm* are standardized. Each term is explained below, with a reference to the section that provides more specific instructions for how to apply it.

Analysis area: This is the area for which daylight calculations are performed. Each analysis area is a distinct room or space in a building, i.e., an area that has a unique, or independent, illuminance requirement. A given building can consist of many, even hundreds, of analysis areas. **Section 4.1** provides a set of rules for determining the analysis area dimensions, what kinds of areas can be excluded or subdivided, and how to lay out a grid of sensor points across that area.

Daylight illuminance threshold value: This is an illuminance level selected to evaluate daylight sufficiency, such that all hourly illumination levels above this threshold count toward the sDA value. The threshold value applied in the calculation of sDA is either 150 lux (15 fc) or 300 lux (30 fc), as described in **Section 4.2**.

In the 2012 edition of this document¹ there was only one illuminance threshold value, 300 lux (30 fc), based on findings from the original research.² However, since publication of the first edition, there has been increasing demand for applying sDA to a greater variety of space types than just those included in the original research. Thus, the committee agreed to define a lower level of

sDA performance based on a 150-lux (15-fc) threshold value, which is appropriate for spaces with lower density of use and lower expectations of illuminance levels, such as some warehouses, lobbies, fitness areas, circulation areas, and support areas. It is recommended that the selection of which of these two values to apply be based on the IES recommended target illuminance criteria for the critical task in a space, or the general illuminance being provided by the installed electric lighting system, whichever is higher, as described in **Section 4.2.1**.

Annual analysis period: This is the daily range of hours used for daylight evaluation for both sDA and ASE. This is standardized at 10 hours per day, between 8 a.m. and 6 p.m. local clock time, for 365 days annually (3,650 hours), following the guidelines in **Section 4.4**.

This 10-hour day was selected as a reasonable approximation of common working hours and because it allows a meaningful comparison across buildings even when they have different operating schedules or when schedules are unknown. Furthermore, the occupancy type of a building may likely change at some time in the future. Thus, sDA is best thought of as describing the asset value of daylight sufficiency in a given space, based on the impact of its enduring architectural design rather than its more variable operational characteristics.

Time fraction: This is the percentage of yearly analysis hours that are considered in the analysis. For sDA, this percentage has been standardized to 50% (1,825 hours), following the guidelines in **Section 4.3.1**. For ASE, it is set at 250 hours (6.85%), as detailed in **Section 4.3.2**. While any value of time fraction could potentially be useful in analysis, these sDA and ASE time fraction values were selected based on findings from the original research, where they provided the greatest ability to differentiate performance across the various study spaces.

Blinds operation algorithm: As mentioned earlier, this algorithm is central to the sDA methodology. For the purpose of analysis, two extreme conditions are first considered, then one with blinds operating is applied. The two extreme conditions are blinds left always open or always completely closed for every day of the year. These two extremes do not necessarily illustrate the best and worst cases, because they do not

account for other aspects of visual comfort, but rather they constitute the brightest and the dimmest cases, i.e., blinds operation that would provide the most and the least direct sunlight and overall daylight that can be admitted (note that horizontal blinds are typically referred to as closed when at a 60-degree slat angle, as they block a view of the horizon in this position). The difference between these two extremes helps define the full range of annual daylight performance that a space can potentially experience, depending on the frequency of blinds application on the daylight apertures.

The sDA with operated blinds is computed for general analysis of the daylight sufficiency within a space and for comparison across different designs. The standardized blinds-operation algorithm for sDA employs what is called a *two-percent trigger*. This blinds operation mode was selected as an approximation of a highly motivated occupant who closes blinds to limit direct sunlight on the work plane. The algorithm closes blinds to prevent sunlight from striking more than 2% of the analysis points in a space at any hour of the year (See **Section 4.6** for full details). There are no other behavioral elements considered other than blockage of direct sunlight; thus, it is understood that this default algorithm has been greatly simplified from operational reality. The goal was to develop and apply a simple standardized algorithm for comparison across all spaces, regardless of room geometry. The two-percent trigger was used in the foundational research to evaluate the interpretation of daylight sufficiency in different space designs, and thus it provides an important benchmark for evaluating spaces.

Since the adoption of the two-percent trigger in the previous edition of this document, a few modifications to its calculation have been made, such as allowing the exclusion of certain non-critical task areas, as described in **Section 4.1.1**. Alternatively, there is also a provision for using automated operation of planned window management devices, as described in **Section 4.6**, if there is convincing evidence that a fully automated daylight management system will be installed in the space. If so, the simulation may use an operation algorithm provided by the automated system control manufacturer or as specified in the construction documents.

To properly evaluate the daylight performance of a given space, all three types of blinds operation should be reported: the two extremes—blinds always open and blinds always closed—plus blinds operating with either the two-percent trigger or an automated algorithm. Each of these should be clearly labeled. If an automated algorithm is used on some or all daylight apertures, the operating assumptions should be clearly documented in the summary report (see **Section 4.15.2**).

2.2 Reporting and Interpretation of sDA

Spatial Daylight Autonomy is intended to be reported using the standardized defaults described in **Section 2.1**, or else with explicit annotation of variations. A subscript of 150 or 300 is used to indicate which default illuminance threshold was used in the analysis. Thus, “sDA_{150/50%}” indicates an analysis using 150 lux (15 fc) as the threshold value, with all other inputs using the standardized defaults. An analysis report for a given space might look like:

sDA_{150/50%}, blinds all open: 92%
 sDA_{150/50%}, blinds all closed: 35%
 sDA_{150/50%}, blinds operated: 78%

Any other variations, such as the use of automated blinds or a different annual analysis period, need to be explicitly stated to assist interpretation.

Based on the surveys of occupants using the study spaces in the foundational research, and a small group of daylighting experts who visited these spaces, the following interpretation of sDA values (with blinds operated) are provided:

- sDA values equal to or greater than 75% are judged to be “preferred”
- sDA values between 55% and 75% are judged to be “nominally acceptable”

3.0 Annual Sunlight Exposure (ASE)

ASE is a metric that describes the risk of visual discomfort from direct sunlight penetration into an interior space. Using a similar simulation methodology to sDA, it

describes the percent of the analysis area within a space that could be subject to an overly bright view of the sun over the course of a year, without the application of daylight management devices (or ‘blinds’ as a shorthand) to limit penetration of sunlight into that space.

ASE is now reported two different ways. The gross ASE considers all image-preserving glazing, while the net ASE considers only the image-preserving glazing areas that do not apply an automated daylight management system (i.e., apertures with, for example, automated blind systems or dynamic glazing are not considered in the calculation of net ASE). If there are no automated daylight management systems at the daylight apertures in a building, then the gross and net ASE values are the same.

ASE is one measure of how dependent a space will be upon blinds operation to maintain visual comfort, given its architectural design. Like sDA, ASE is intended to provide an assessment of the asset value of the space, by accounting for its fixed characteristics that influence sunlight penetration, such as building and aperture geometry, glazing features, exterior obstructions, and climatic location. ASE is not intended to predict actual blinds operation once the space is occupied because that is influenced by many additional and less permanent factors, such as the layout of furniture, the type of window coverings selected, their control devices, and, of course, the specific needs and behavior of the occupants in the space, all of which may vary considerably over the life of the building.

Field research² found that the more frequently the work plane could experience direct sunlight across the year, and the larger the area that could receive direct sunlight, the more likely the occupants were to be dissatisfied with the visual comfort conditions of their daylit workspace over the course of a year. It is important to note that these occupant observations were made relative to whatever operable blinds existed and were functioning within each space over the course of the previous year.

ASE values can be used to gauge the need for mitigation strategies to reduce potential sources of glare or visual

discomfort from direct sunlight. However, it is important to note that ASE is simply a first-order level of analysis for potential glare, and it does not consider all possible reasons that occupants might choose to operate their blinds. In addition to a direct view of the orb of the sun, other glary conditions could include excessive contrast or troubling reflections resulting from patches of direct sunlight on interior or exterior surfaces, the glint of sunlight off window frames or other shiny surfaces, or high luminances that might be delivered by the selected daylight management devices or diffuse glazing. Thus, ASE should only be used as an indicator of potential visual discomfort from direct sunlight penetration through windows and skylights, not as a way to prove visual comfort, or to identify glare.

3.1 Standardized Inputs, Terms, and Concepts

ASE is defined as the percentage of an analysis area that would receive more than 250 hours of *direct sunlight* per year, *before* blinds are deployed.

For the purposes of the ASE calculation, *direct sunlight* is defined as at least 1,000 lux (100 fc) of sunlight beam illuminance greater than the ambient diffuse daylight levels inside a space on a horizontal plane (see additional descriptive information below). This is approximately the threshold level of sunlight that will cause noticeable shadows, or when the orb of the sun is first visible at the horizon or through a layer of clouds. It can be interpreted as the difference between the horizontal illuminance in a patch of sunlight and the illuminance in a directly adjacent shadow patch (not in direct sunlight).

A threshold value of 1,000 lux (100 fc) was selected to represent the presence of direct sunlight (as defined in **Section 4.2.2**) and to answer the questions “Is the sun out?” and “Could a view of the orb or resulting interior sun patches be a source of visual discomfort?” While there are many other possible ways to identify visual discomfort from direct sunlight, such as measurements of luminance (in nits or candelas per square meter, cd/m²) or energy intensity (such as watts/m²), illuminance was selected for the purpose of determining ASE because it can be identified from the weather file, is easy to compute, and is required for the calculation of sDA. **Section 4.2.2** explains how to identify such direct

sunlight conditions using TMY weather files in an annual simulation.

Direct sunlight: This is the unscattered portion of a sunlight beam arriving at a point after passing through image-preserving glazing material in a daylight aperture. This includes beam penetration through any daylight aperture, such as a window, glazed door, clerestory, or skylight. For a specific sky condition, it accounts for any exterior obstructions (including fixed shading), glazing transmittance, the angle of incidence on the glazing, the impact of mullions, and any applicable dirt depreciation multiplier, but it excludes the effect of any blinds or operable shading devices. The ASE calculation contains no contributions from reflected light (i.e., it assumes a zero-bounce analysis) and no contribution from the diffuse-sky component.

ASE analysis uses the same daily 10-hour period of analysis as sDA uses, as described in **Section 4.4**. However, the position of the sun in the sky and the intensity of its light can vary considerably over the course of an hour. For the purposes of the ASE simulation, the 1,000-lux (100-fc) value is derived from a TMY weather file based on a one-hour average. Thus, it is understood that sunlight intensity is likely to be greater than this value for some portion of that one-hour period. Another simplification employed in the calculation is the position of the sun, which is determined by the solar position at the center of each hour-long period, i.e., on the half-hour (e.g., 8:30, 9:30).

As with sDA, an ASE calculation standardizes the time component and reports out the resulting area that exceeds a criterion level. Based on analysis of expert and occupant evaluations of the same 61 spaces used to develop sDA, 250 hours of sunlight per year was selected as the threshold amount of time that best differentiates which points within a space could be problematic compared to those that are more acceptable.

3.2 Reporting and Interpretation of ASE

An ASE analysis reports the percentage of analysis points within each space that could exceed 250 hours of direct sunlight per year if no blinds were operated to reduce direct sunlight. Two different ASE values are reported: first, across all image-preserving glazing (ASE_{gross}); and

second, across all image-preserving glazing that lacks automated glare control (ASE_{net}). ASE is intended to be reported only using the standardized defaults and methodology described in this document, or else with explicit annotation of variations.

Furthermore, any area that reports sDA should also report ASE across the same analysis area. ASE should be considered a relative value, where a smaller area is understood to be preferred, and progressively larger areas are more worrisome, from the perspective of the potential to produce glare. Thus, for a south-facing space, ASE_{net} would likely be fairly high, perhaps around 30%:

$sDA_{150/50\%}$, blinds all open: 92%; $ASE_{net(1000/250h)}$: 30%

$sDA_{150/50\%}$, blinds all closed: 35%

$sDA_{150/50\%}$, blinds operated: 78%

Based on experience to date and re-analysis of the study data, an ASE_{net} of 20% or higher has been determined to be a level of concern. (This value was listed as 10% in the 2012 version of this document.) For spaces where ASE_{net} exceeds 20%, it is suggested that the designer consider architectural methods to reduce the risk of sunlight penetration, or consider the use of an automated daylight management system in as many window areas necessary to guarantee that excessive occurrence of direct sunlight can be avoided within the space.

ASE values are specific to each space evaluated. They do not make sense if aggregated or averaged across multiple analysis areas or across an entire building. Rather, ASE values are used to qualify spaces as achieving sufficient daylight illumination without the presence of excessive sunlight. For this reason, only those spaces where the ASE_{net} is less than 20%, with glazing areas that contain specified automated daylight management devices excluded from the analysis, should be included when assessing sDA for an entire building. This requirement to maintain ASE_{net} below 20% encourages glare mitigation technologies as well as improved fenestration and envelope design to limit solar exposure.

4.0 Annual Climate-Based Daylight Modeling Methodology

This section describes the simulation methodology to be applied in the analysis of sDA and ASE in architectural spaces. It describes the precision of architectural details, the layout of analysis-area calculation points, illuminance threshold values, time fractions, annual analysis period, annual blind operation, and other simulation specifications—such as default reflectance values, window blind and shade transmittance, and outdoor geometry—required to produce values that are compliant with this document.

In general, spaces should be simulated as realistically as possible, based on verifiable values from construction documents or onsite measurements. It is critical that the geometry of the space be accurately modeled, including angled walls or ceilings, as well as all fenestration surfaces and aperture conditions, including exterior-wall thickness and the appropriate orientation of the apertures with respect to true solar north. If important parameters are not known, LM-83 default values (specified in the subsections that follow) should be used. Many analysis parameters are fixed in order to report a standardized daylight asset value that can be compared across buildings, analysis areas, and design alternatives. The parameters listed in **Sections 4.1** through **4.11** are explicitly required to follow the LM-83 methodology, and any alternate assumptions or methods that were applied are to be explicitly noted with those results, such as in customized investigations that may be carried out during the design process.

4.1 Analysis Area

Spatial Daylight Autonomy measures daylight sufficiency across a contiguous space based on a grid of *analysis points* that define an *analysis area*. The output is presented as the percentage of analysis points in an analysis area that meet the stated criteria (e.g., those that exceed 300 lux (30 fc) for at least 50% of the analysis period from 8 a.m. to 6 p.m.). The same analysis areas and analysis locations used to calculate sDA shall be applied to calculate ASE, which is to be reported along with the sDA values.

4.1.1 Excluded Areas. Each analysis area shall be assigned in the following manner:

- A building is first subdivided into individual spaces (such as rooms), which are defined by volumes significantly surrounded by solid surfaces, or in the case of an open floor plate, by window orientation.
- Areas with a space where the installed lighting systems provide different illuminance levels that place them under different sDA threshold values as outlined in **Section 4.2.1** should be addressed as separate analysis areas.
- Spaces are then assigned a single analysis area that is articulated with a set of analysis points uniformly distributed across the space (following the criteria described in **Section 4.1.2**).

Spaces with identifiable boundaries that are associated with “non-critical visual tasks” (such as circulation areas, restrooms, and non-active storage areas) may be included or excluded from the analysis area at the discretion of the design team. However, these places are to be included or excluded consistently for analysis of both sDA and ASE across the building and across similar spaces.

In the case of a space that is designed with a sunlight-tolerant area, and which is clearly indicated as a design feature in the architectural plan, this area may be excluded from the daylight analysis. An example sunlight-tolerant area is a perimeter sunlight buffer band adjacent to the windows, or an interior atrium for growing plants. Documentation is required to ensure that all areas in a sunlight-tolerant area exclude task areas such as desks and workstations. Core-and-shell projects may only exclude analysis areas that are physically differentiated from other spaces, as with permanent partitions or a change of floor elevation.

4.1.2 Analysis Points. Analysis points shall be continuous and specified no greater than 0.61 m (2 ft) on center across a space or study area. To lay out an analysis grid across a space, the grid may be located with any starting point, such that the entire space or area is covered, wall to wall in all directions, and then points within 0.30 m (1 ft) of the room walls or other space boundary may be excluded from the analysis.

Points are typically placed at a height of 0.76 m (2.5 ft) above finished floor, or 2.5 cm (1 in.) above the highest regular task level. The analysis point grids are preferably centered in a space. Any analysis points that are inside furniture, such as within the interior of a bookcase or file cabinet, may be dropped from the analysis.

4.2 Illuminance Threshold Value

4.2.1 sDA. For the calculation of sDA, the illuminance threshold value assigned is either 300 lux (30 footcandles) or 150 lux (15 fc), measured at the horizontal work plane, which by default should be 0.76 m (2.5 ft) above the finished floor, unless more-specific information is known. The appropriate threshold value is assigned to each analysis area according to the guidelines provided below or based on alternate requirements provided by a particular jurisdiction, code body, or green-building rating organization.

A threshold value of 300 lux (30 fc) is used to calculate $sDA_{300/50\%}$ and is intended for the analysis of daylight sufficiency in spaces where 25% or more of the analysis area meets either one of the following criteria:

- It contains visual tasks that align with an *IES horizontal ambient illuminance design target* of 225 (22.5 fc) lux or greater, as specified in the relevant IES Recommended Practice document.
- The work plane illuminance provided by the designed electric lighting system for the space (after lumen maintenance and dirt depreciation factors are applied) meets or exceeds 225 lux (22.5 fc).

A threshold value of 150 lx (15 fc) may be used to calculate $sDA_{150/50\%}$ when neither of the above criteria is met.

In cases where a non-standard threshold value is applied to an sDA study (for example, to support design analysis), the threshold value should be explicitly stated, as in “ $sDA_{500/50\%}$.”

4.2.2 ASE. The illuminance threshold value for ASE is 1,000 lux (100 fc), which is for direct-beam sunlight that passes through image-preserving glazing, or through glazing material that has an image-preserving

component, such as fritted glass. When the glazing material transmits daylight through a combination of direct-beam and diffuse regions across its surface, only the unobstructed and undiffused direct-beam sunlight is considered in the ASE analysis. In addition, it is important that the simulation method applied maintain the parallel-ray nature of any directly transmitted sunbeam (i.e., the use of BSDFs for clear glazing may not provide acceptable results for the analysis of the direct sunlight contribution required for both the sDA trigger and ASE). ASE is computed and reported two different ways: first, with blinds “all open” and with dynamic image-preserving glazing at maximum transmittance (this value is referred to as ASE_{gross}); and second, with all daylight apertures that contain fully automated shading devices, including dynamic glazing, excluded from the ASE analysis (ASE_{net}).

4.3 Time Fraction

4.3.1 sDA. The time fraction is the percentage of the analysis period (study hours per year) that a specific threshold value is required to be met. Spatial Daylight Autonomy is reported as the percentage of an analysis area that achieves the relevant illuminance threshold value for 50% of the analysis period (8 a.m. to 6 p.m., 365 days per year). Based on this 10-hour annual analysis period, 50% represents 1,825 hours per year. If an alternate time fraction is used (for example, to support design analysis), it shall be explicitly specified (e.g., sDA_{300/70%} indicates that a 70% time fraction has been applied).

4.3.2 ASE. The time fraction for ASE is specified by a set number of hours, namely 250 hours, that an analysis point receives 1,000 lux (100 fc) or more of direct sunlight across the analysis period. If an alternate time fraction is used for an ASE study (for example, to support design analysis), it shall be explicitly specified (e.g., ASE_{1000/100h}).

4.4 Annual Analysis Period

As described in **Section 2.1**, the period of analysis for sDA and ASE is fixed at 10 hours per day, from 8 a.m. to 6 p.m. local clock time, accounting for daylight saving time if it applies at the building location. This results in 3,650 hours for a complete annual analysis.

Daylight availability varies dramatically by time of day, as do climate conditions in many locales, where clouds or fog may be more frequent in morning or afternoon hours. Thus, changing the analysis period will inevitably also change annual daylight availability, and therefore the output of any analysis. While other periods of analysis may be useful for internal studies to optimize a given design, or for research purposes, a fixed period is required to set widely shared standards of performance. If an alternate analysis period is used to conduct a non-standard sDA study, that analysis period shall be explicitly specified (e.g., sDA_{300/50%, 8am-5pm} and ASE_{225, 8am-5pm}, where the sunlight threshold hours have also been proportionally reduced to account for the shorter analysis period).

Formatted files containing TMY hourly weather data that are applied in daylighting analysis involve hourly averages, with data centered on the half-hour. For example, Hour 9 on a particular day is an average for the 8 a.m. to 9 a.m. hour; therefore, for the modeling of sDA and ASE, this time band should consider the solar position associated with 8:30 a.m. All other hours of the day are to be considered in a similar manner.

4.5 Windows and Skylight Details

Window and skylight glazing shall be modeled with actual visible light transmittance (VLT), less a dirt depreciation fraction (i.e., the glazing VLT is multiplied by a factor that is equal to 1.0 minus the dirt depreciation fraction expressed as a decimal). Window and skylight glazing materials should be modeled as clear (specular; non-diffusing) and/or diffusing (Lambertian), consistent with their optical properties, unless more-precise optical performance data are known and can be specified in the model using bidirectional scattering distribution function (BSDF) data.⁴ Materials such as fritted glass or roller shades with an openness factor, which have both a specular and diffuse transmission component, shall be modeled as an appropriate combination of both specular and diffuse transmission.

Window and skylight openings shall be modeled in three dimensions, as described in this section. When not known, other framing elements, such as mullions, shall be modeled as a percentage reduction in glazing VLT relative to a rough opening area, as described below. For

skylights, the vertical offset between skylight glazing and ceiling surface, otherwise known as the skylight well, shall be modeled as designed or constructed.

- Fenestration glazing VLT shall be based on site-measured data for existing buildings, or manufacturer's product information for new construction.
 - Dirt depreciation factors shall be applied to the specified VLT. At a minimum, 5% dirt depreciation shall be subtracted from the VLT of vertical or exterior downward-facing windows through the application of a 0.95 multiplier (the dirt depreciation factor); 10% dirt depreciation shall be subtracted from the VLT of glazing that is sloped between 20 and 85 degrees from horizontal (a 0.9 multiplier); and 15% dirt depreciation (a 0.85 multiplier) is applied to the VLT of skylight surfaces that have less than a 20-degree slope, following Hopkinson et al.⁵ Higher dirt depreciation values may be appropriate for very dusty areas, or for buildings with poor maintenance.
- Any window detail (e.g., sills, jambs, mullions) greater than 5 cm (2 in.) in any dimension shall be modeled as such.
 - Alternatively, the area of the sills, jambs, and mullions normal to the glazing surface can be calculated and summed into an "occlusion area." The ratio of the occluded area to the area of the rough opening across which the glazing is applied can then be used as a reduction fraction $[(\text{occlusion area})/(\text{gross window area}) = \text{mullion reduction fraction}]$ to account for the opaque window elements. The gross window area is then simulated with both this reduction fraction and the dirt depreciation fraction applied to the VLT value for the glazing. For example, if the gross window area or rough opening is equal to 10 square meters and the occluded area is 1 square meter, a 10% mullion reduction fraction can be applied to the VLT of the glass in place of a full geometric model of the mullions. In combination with dirt depreciation, vertical glazing with a 70% VLT would be simulated with a VLT of $0.70 (1 - 0.1) (1 - 0.05) = 0.60$.
 - When window-framing details are not known, a 20% mullion reduction fraction should be applied

to the glazing VLT assigned to the rough opening. For skylights, a 10% reduction fraction should be applied when skylight-framing details are not known and the glazing material is applied across the rough opening.

- All overhangs, light shelves, and other light redirecting elements shall be modeled with accurate geometry and material transmission and/or reflection properties.
- All window openings shall be modeled with actual wall thicknesses and sill and jamb reflectances.
 - When not known, window recesses, jambs, and sill depths should be assumed to be 15 cm (about 6 in.) on either side of the glass surface and modeled with a 50% reflectance.
- All skylights shall be modeled as three-dimensional elements, using actual or specified dimensions, geometry, VLT, and reflectance.
 - If known, BSDF files for optically complex skylights may be used.
 - Otherwise, diffusing skylights should be modeled with a Lambertian distribution, clear skylights should be modeled as transparent, and materials such as fritted glazing should be modeled using an appropriate combination of diffuse and transparent transmission.
- All image-preserving glazing, regardless of its location (skylights included), shall be modeled with blinds that operate as part of a window group to limit sunlight penetration, as outlined in **Section 4.6**. All image-preserving glazing is also considered in the calculation of ASE within a space. An exception for automated electrochromic glazing is provided in **Section 4.8**.
- Interior or exterior blinds or other shading elements shall be modeled per the discussion on blinds operation and optical properties in **Sections 4.6** and **4.7**.
- The fenestration surfaces shall be properly oriented with respect to vertical and true solar north within the model.

4.6 Blinds Operation and Window Groups

For sDA calculations, all exterior windows and skylights that are image preserving shall be modeled with interior

blinds that close by *window group*, even if no blinds have been specified in the design. A window group is a collection of coplanar, or nearly coplanar, windows with similar shadow patterns from exterior shading and obstructions, with similar blinds type and operation, or similar glazing material, that are associated with the same analysis area. The properties of the blinds are those for what is expected to be installed in the real space, or the LM-83 default 5% diffuse transmittance, as stated in **Section 4.7**. The default value for blinds transmission is also applied on image-preserving glazing where no blinds have been specified. Window groups that contain glazing with diffusing properties that eliminate the direct sunlight beam, or those with glazing with adjustable transmittance that may be reduced to less than 2%, do not require interior blinds.

For the simulation of sDA, the granularity of window groups within a space may range from the incorporation of all glazing area on one exterior wall of a space into a single window group to the separate modeling of individual windows, provided that no blind operating unit is subdivided; that is, windowpanes protected by the same blinds unit shall not be assigned to different window groups. Thus, a room with three manually controlled blinds would be modeled with no more than three window groups. For automated blinds, window groups are based on sections of daylight apertures that are controlled together. In spaces with complex exterior wall geometry, windows within ± 10 degrees of the same orientation may be included within the same window group; however, finer angular separation between groups is permitted.

The control of blinds on different window groups in an sDA analysis shall apply a 2% trigger, unless an automated daylight management system is present, in which case the specified control algorithm for this system shall be applied. Within each space, the control of blinds on all window groups that operate under the standard 2% trigger shall limit direct sunlight of 1,000 lux (100 fc) or higher to no more than 2% of the analysis points in the space at every hour of the year. Any combination of fully open and fully closed settings across the different window groups that meets the 2% rule is permitted. The tilt angle of louvered blinds is adjusted each month to be as open as possible while

excluding direct-beam daylight at all hours of the month across the 10-hour study period (see **Section 4.7**).

When automated control of blinds is applied, the automated control algorithm may include partial closure of blinds within a window group (i.e., some systems might lower a blind to cover only a fraction of the window under certain daylight conditions, such as to limit sunlight penetration depth). Under these conditions, for the purpose of determining the blind closure conditions for glazing areas that have blinds that are controlled by the 2% trigger rule, only sunlight penetration through the non-automated daylight control fenestration areas is considered.

In the recommended formal reporting of sDA values for spaces (see **Section 4.15**), three different blinds-operation conditions need to be considered:

- Blinds all open
- Blinds all closed
- And one of the following:
 - 2% trigger
 - Automated
 - Combined (see “Combined” bullet below)

These reporting conditions are defined as follows:

- *Blinds all open*: Blinds are not operated, and image-preserving glazing is unobstructed every hour of the year.
- *Blinds all closed*: Blinds on image-preserving glazing are always closed (i.e., the glazing is fully obstructed every hour of the year, whether control is via the 2% trigger or automated). When blinds are applied to a window group, they shall completely cover the daylight apertures. In the case of dynamic glazing, a 2% transmittance is applied for this reporting condition.
- *2% trigger*: Blinds are operated hourly to prevent direct sunlight from striking more than 2% of the analysis points in an analysis area during any given hour of the year. The hourly direct-beam sunlight values from the ASE simulation are applied to assess the percentage of the analysis points that receive direct sunlight.

- *Automated:* Automated blinds or another dynamic daylight management system that automatically eliminates or mitigates direct sunlight penetration shall be operated as specified in the construction documents. A description of the blinds operation specifications shall be provided. Partial occlusion of a daylight aperture is permitted with an automated blinds-control algorithm.
- *Combined:* In this case, window groups with both automated control and a 2% trigger are present within a space. The automated control is applied as specified in the construction documents, while all other window groups are controlled using a 2% trigger. However, any sunlight penetration through window groups with automated daylight management system is excluded from the 2% sunlight trigger analysis.

4.7 Optical Properties of Blinds and Shades

To define the combination of specular and diffuse transmission of any blind or shade material, BSDF files, special material definitions, or a representative geometric model (e.g., of louver blinds) should be used, if available. For modeling purposes, the angle of adjustable slatted blinds shall be fixed and reset monthly at an angle that completely blocks direct sunlight from the lowest solar angle experienced on that facade orientation in each month at that location across the study, based on the nearest TMY weather file for the 10-hour analysis period.

A permissible alternative to the application of BSDF data in the case of fabric shades, curtains, or louvered blinds is for the shades to be modeled using a combination of specular and diffuse transmittance and an appropriate reflectance. The *specular transmittance* should be equal to the *openness factor* of the fabric,⁶ while the *diffuse transmittance* should be the total VLT minus the openness factor. If the VLT is known and the openness factor is not known, the VLT should be modeled as diffuse transmittance only. If the shade VLT is unknown, the shade should be modeled using 5% diffuse VLT with no specular transmittance.

In the case of louvered blinds, a simplified, permissible alternative to either applying BSDF data or modeling the physical geometry of the angled blinds involves

considering the blinds as a diffusely transmitting (i.e., Lambertian) material as follows. If the louvered blinds material has a reflectance of 80% or higher, a 20% VLT diffuse distribution should be used to assess all daylight transmitted through the blinds. The VLT of darker blinds colors shall be depreciated proportionally, to a lower limit of 10% diffuse VLT for black blinds (reflectance of 0% to 10%) for both sunlight and sky light. Thus, for example, blinds with a 40% reflectance should be modeled at 15% VLT, while a 60% reflectance is modeled at 17.5% VLT. Other examples are shown in **Table 4-1**.

Table 4-1. Examples of Simulated Diffuse VLT Values for Louver Blinds of Various Reflectance Values

Louver Blinds Reflectance	Simulated Diffuse VLT Value
>80%	20%
70%	18.75%
60%	17.5%
50%	16.75%
40%	15%
30%	13.75%
20%	12.5%
<10%	10%

It is important to note that the transmittance values of blinds and shades, when these devices are applied, are to be considered in addition to the VLT of the glazing material, while also accounting for dirt depreciation as specified in **Section 4.5**.

4.8 Dynamic Glazing

For dynamic glazing, such as electrochromic glass, or other automated daylight management systems, the transmittance properties should be adjusted according to the actual algorithm and general layout that will be applied when simulating sDA. If the glass is image preserving at all settings, the appropriate transmittance (including a dirt depreciation factor) is applied to clear glass at each hour in the daylight simulation.

A simplified approach may also be applied. In this case, the 2% trigger is applied across all window groups, and for the electrochromic glass, the application of blinds is replaced with an adjustment of the VLT of the glass down to 2% (before any dirt depreciation

factor is applied). At this setting, the direct sunlight through dynamic glazing is assumed to be negated for the purpose of additional shade-trigger considerations, regardless of its value. This 2% transmittance is also applied to the glazing for the “blinds all closed” sDA analysis. Window groups should be organized for group control as specified in the construction documents when modeling dynamic glazing.

4.9 Exterior Obstructions

Exterior obstructions shall be modeled using at least the minimum level of detail described here:

- All buildings and opaque structures shall be modeled that obstruct a portion of the sky more than 10 degrees above the horizon, as viewed from any point at ground level along the base of the building. Exterior obstructions shall be modeled with a resolution of at least 3-m (10-ft) increments in their dimensions.
 - More-accurate and more-extensive exterior obstruction modeling is permitted.
 - If the reflectance of surfaces is not known, a reflectance of 30% shall be assumed for all vertical exterior surfaces.
 - If exterior obstructions are not known, adjacent buildings of height equal to that of the study building, set on adjoining and opposing parcels with a setback equal to that of the study building, shall be assumed.
- Trees shall be modeled as appropriately sized cones, spheres, or cylinders, with a 20% reflectance. More-accurate shapes are permitted. Assumed tree heights should use the expected height of each tree at least 5 years beyond the date of completion of the proposed project.
- Actual surface reflectance values shall be used for ground surfaces within a minimum of 30 m (100 ft) of the building perimeter. The computer model should be designed to adequately model shadows on the ground within this area. (*Note:* Some tools require an exterior ground polygon to be included adjacent to a building to model these shadows.)
 - If actual reflectance values are not known, a 10% reflectance shall be used for the ground surface.

4.10 Interior Surface Reflectances

Interior surface reflectances shall be modeled based on information from field measurements, construction specifications, or, if unknown, by applying the default values provided below.

When a planar surface contains large areas (greater than 3 m [10 ft] in any dimension), with reflectances varying by more than ± 10 percentage points from the primary-area reflectance, those areas shall be modeled separately. For example, in a classroom, a large whiteboard with a reflectance of 85% would be modeled separately from a wall of 50% reflectance, since it is outside the range of ± 10 percentage points range, which would be 40% to 60% for this wall surface.

If actual reflectance values are not known for walls, floors, ceilings, or furniture, the following default reflectance values shall be used with a gray material:

- Floor: 20%
- Walls: 50%
- Ceiling: 70%
- Furniture: 50%

4.11 Furniture and Partitions

Furniture and interior partitions that meet the following criteria shall be modeled:

- Any partition or furniture extending 0.9 m (3 ft) or more above the floor shall be modeled to within an accuracy of 0.2 m (0.5 ft). Simplified mass objects are acceptable. If furniture layout and type are not known precisely, a typical furniture layout for that space type should be used.
- Any interior glazing on partitions shall be modeled using the VLT of the glazing material, along with its transparency and diffusion characteristics.

4.12 Climate Modeling Methodology

This section describes the necessary simulation methodology for the site climate conditions in the calculation of sDA and ASE. It describes the precision of climate modeling, including solar position, illuminance due to sky and sun, and accuracy of various modeling inputs.

Simulations are run using sky condition inputs based on TMY data. A TMY weather data file, or equivalent, is

selected from the nearest available weather station that approximates the climatic and geographic conditions of the study site. The TMY data is used to derive the solar luminous intensity, sky-dome luminance distribution, and probability of typical weather patterns over the course of a full year. Any deviation in weather file type should be noted in the reported simulation results.

4.12.1 Solar Position and Intensity. The angle between the simulated solar position and the actual solar position for all times and locations should be less than 1 degree when computing the direct solar beam contribution. Preferably, exact sun positions on the half-hour, using the altitude and azimuth locations based on the calculated position of the sun, should be applied for ASE and sDA analysis. If the sky dome is divided into equally sized or spaced patches for addressing the direct solar contribution, this results in no fewer than 2,305 sky patches.

Hourly solar direct normal illuminance should be modeled using TMY data and the Perez All-Weather Sky Model.⁷

The reflected contribution from the sunlight beam can be modeled by applying a sunlight contribution to the sky patches used for the sky contribution, provided that the direct sunbeam contribution is replaced with one of the more refined methods described in this section if fewer than 2,305 sky patches are applied.

4.12.2 Sky Luminance and Distribution. The sky should be modeled with a minimum of 145 sky patches, per Reinhart,⁸ if daylight coefficients and/or discrete sky patches are applied.

The hourly diffuse sky component should be modeled using TMY weather data and the Perez All-Weather Sky Model.⁷

4.13 Modeling Parameters

Modeling parameters are complex and vary according to analysis program algorithms and the features of the daylit space. Simulation software should be selected that is capable of accurately modeling the spatial geometry and daylighting technologies considered. The modeler shall select sufficiently high simulation parameters

to yield reliable results, ideally based on sensitivity studies (guidance is available⁹). Modelers are advised to consult with experts in the use of the software package they are employing. The primary types of program algorithms used in daylighting are ray tracing and radiosity. Calculation modeling parameters should be set to resolve any sharp gradients in luminance, as well as to properly address important reflecting elements, such as windowsills, light shelves, and louvers.

Image-preserving glazing should have a transmittance that varies with incident angle, either applying WINDOW 7 data or using a standard transmission curve for uncoated glass that addresses the appropriate number of glass layers present.

Radiosity calculations should address the window as a series of light emitting elements no larger than 0.3 m x 0.3 m (1 ft x 1 ft). Room-interior light-emitting elements should be no larger than 0.6 m x 0.6 m (2 ft x 2 ft), with light-receiving elements preferably no larger than 0.3 m x 0.3 m (1 ft x 1 ft) where there is a strong luminance gradient. Larger window elements may need to be subdivided if they have different shadowing patterns or incident-light distribution.

It is important that software used to calculate sDA can implement the 2% sunlight-trigger blinds algorithm specified in this document, or an automated algorithm specified by the designer or shade manufacturer, for sDA results to comply with this document.

4.14 Simulation Output Formats

Simulation output formats should enable users to examine geometric plots of the data used to derive sDA and ASE (i.e., plots of daylight autonomy values used for the sDA calculations and the annual sunlight hours received at each analysis point in the ASE simulations) and indicate space orientation relative to true solar north. Plots should also clearly indicate the applicable analysis areas. The ability to examine geometric plots with hourly or monthly summaries is recommended but not required.

Annual summary values for each identified analysis area shall be reported for both sDA and ASE so that they can be considered together. Summaries should include

sDA for fully open, fully closed, and blinds-operated, as described in this document. Output reports should also indicate the number of window groups per analysis area. The number of hours that blinds on each window group are activated for the operable blind case is also useful information.

When the analysis is for an entire building, the report should note those areas that are excluded from the analysis of individual spaces, and the resulting percentage of gross building area that has been excluded.

All sDA results shall indicate the type of blinds operation applied and shall provide a description of the algorithm for any automated system.

Justification for any variation from standard modeling methodology, as described in this document, shall be noted.

4.15 Summary Reports

When reporting sDA scores for the purpose of compliance with this document, tabulated data on the performance in each space, across each facade, and across the entire building, as well as detailed information on the glazing and shading device properties, should be provided.

Recommended formats are provided in the subsections that follow.

4.15.1 Space-by-Space Daylight Simulation Report.

An sDA table containing the following information for each space shall be provided; **Table 4-2** provides an example. (Note that spaces that have no daylight but are included in the building summary are also listed.)

- Space name (or room number)
- Orientation grouping
- Space type
- Analysis grid area
- sDA for blinds always open
- sDA for blinds always closed
- sDA for blinds operated (e.g., results that apply either a 2% trigger or an automated-blinds algorithm, or some combination (whichever applies)
- ASE_{gross}, the ASE value for all image-preserving glazing, including apertures with automated control.
- ASE_{net}, the ASE value for apertures with image-preserving glazing and blinds operating using the 2% trigger, but excluding all apertures that have automated systems, such as blinds or dynamic glazing.
- A list of the IDs (names) for each window group in each space.

Table 4-2. Example of a Space-by-Space Daylight Simulation Report

Space Data				Annual Daylight Metrics						
Space Name	Space Orientation	Space Type	Analysis Grid Area* (ft ²)	sDA Threshold Value (lux) 300 or 150	sDA No Blinds (%)	sDA Blinds Down (%)	sDA Blinds Operating (%)	ASE gross (%)	ASE net (%)	Window Group ID
101	90	Lobby	300	150	80.0	20.0	60.0	12.0	12.0	101E
										101S
102	270	Office	150	300	75	25	60.0	8.5	0.0	102W
103	270	Office	150	300	70	25	55.0	13.3	0.0	
104	270	Office	150	300	70	20	55.0#	22.0	22.0	104W
105	0	Office	600	300	70	20	68.0	0.0	0.0	North
106	Interior	Office	200	300	0	0	0.0	0.0	0.0	
107	Toplight	Office	600	300	70.0	20.0	68.0	8.0	8.0	SkyIt

A value of 0 is used for this space in the Building Summary since ASE is greater than 20% and blinds are not automated.

* Areas can be m² or ft².

The sDA value with blinds operated based on annual hourly sunlight penetration shall be used for any compliance criteria. If neither a 2% blinds trigger nor an automated-blinds sDA value is provided, then the blinds-closed sDA results are applied. The blinds-operation sDA table is useful in comparing design alternatives and for review by jurisdictions and code and rating organizations to ensure simulation protocols have been correctly implemented.

4.15.2 Space-by-Space Window Group and Glazing Report. For each space, a Window Group Report (an example is shown in **Table 4-3**) is recommended to provide material and performance details on each of the window groups with the following recommended content.

- Window group ID (i.e., name)
- Glazing area
- Glazing transmittance (this value should be the manufacturer's listed value, but in the analysis, a dirt depreciation factor and, if needed, a mullion reduction factor shall also be applied to the glazing VLT)
- Blinds ID
- Blinds control method

- Hours blinds are down (for 2% trigger)
- Hours with >50% occlusion (for automatic control, if applied)

For electrochromic (EC) glazing (if present), the following shall be reported:

- Window group ID (i.e., name)
- Glazing area
- Operation method
- Hours VLT is less than 10%

4.15.3 Schedule for Sunlight Glare Control Devices. A schedule that documents the properties of shading devices to control sunlight glare (**Table 4-4** shows an example) should accompany the Space-by-Space Daylight Simulation Report table. This schedule includes transmittance data and a product description for each of the blinds, diffuse glazing, and EC-glazing products that have been listed in the Space-by-Space table. It also includes space for a general description of the automated control method that was applied to blinds and EC glazing.

Table 4-3. Example of a Space-by-Space Window Group and Glazing Report

Space Data			Glazing (Clear and Diffuse) with Blinds Data						EC Glazing			
Space Name	Space Orientation	Space Type	Window Group ID	Glazing Area (ft ²)	Blinds ID*	Blinds Operation Method	Hours Blinds are Down (2% Trig)	Hours >50% Occlusion (Auto)	Window Group ID	Glazing Area (ft ^{2#})	Operation Method	Hours VLT ≤ 10%
101	90	Lobby	101E	50	RS1	2% Trigger	1,200	--				
			101S	50	Diffuse Glazing	--	--	--				
102	270	Office	102W	50	RS2	Automated	--	500				
103	270	Office							West-EC	100	EC Glazing	250
104	270	Office	104W	50	BL1	2% Trigger	500					
105	0	Office	North	80.0	Diffuse Glazing	--	--	--				
106	Interior	Office										
107	Toplight	Office	Skyt	30.0	RS1	2% Trigger	1,000					

* Diffuse glazing does not require blinds for sDA simulations.

Areas can be m² or ft².

Table 4-4. Example of a Schedule for Sunlight Glare Control Devices

BLINDS ID	TYPE	Operation Method	VLT*	Description (e.g., With Performance Values, Openness, Electrochromic Dimming Levels)
RS1	Roller Shade	Manual	0.10	Supershade with Tvis=0.1 and 2% openness factor, light cream
RS2	Vertical	Automated	0.05	Gray shades controlled by photosensor and astronomic timeclock
SPECIAL GLAZING				
Diffuse Glazing	--	--	0.30	Other product details go here
EC Glazing	--	--	0.01	Other product details go here

* For electrochromic glazing, this should be the minimum transmittance possible. For interior blinds, VLT is for the blind alone.

4.15.4 Building Daylight Summary Reports. An example of a Building Daylight Summary Table is shown in **Table 4-5**, which provides compiled sDA and ASE data over the different facade orientations, interior area, and skylight area, as well as a full-building summary of sDA and ASE. An accompanying summary of the fraction of the gross building area that has been assigned to the two sDA threshold categories as well as the areas excluded from sDA and ASE analyses—sunlight-

tolerant buffer areas and the core and non-critical task areas—are also included. It is worth noting that the Daylight Summary Table lists analysis grid areas, which are typically smaller than the space area to which they are assigned due to the distance from the first and last point in a row or column to the wall. The percentages of the gross building area that are listed in the Building Floor Area Daylight Category Summary are derived from the gross area of each of the spaces.

Table 4-5. Example of a Building Daylight Summary Report

Orientation	Analysis Grid Area (ft ²)*	sDA No Blinds (%)	sDA Blinds Down (%)	sDA Blinds Operating (%)	ASE gross (%)	ASE net (%)	Glazing Area (ft ²)*	Blinds Operation Method	Average Annual Blind Occlusion (%)	Diffuse Glazing Area (ft ²)*	EC Glazing Area (ft ²)*	Annual Area Averaged VLT*
90	300	80.0	20.0	60.0	12.0	12.0	50	2% Trigger	32.9%	50		
270	450	71.7	23.3	38.3	14.6	7.4	100	Multiple			100	0.41
0	600	70.0	20.0	68.0	0.0	0.0				80		
Interior	200	0.0	0.0	0.0	0.0	0.0	0					
Toplight	600	55.8	20.0	68.0	8.0	8.0	30		27.4%			
TOTAL	2,150	65.2	16.1	54.3	7.0	5.5	130		30.8%	130	100	0.41
BUILDING GROSS FLOOR AREA BY DAYLIGHT CATEGORY SUMMARY												
300 lux Threshold							45%					
150 lux Threshold							22%					
Sun-tolerant area							7%					
Core and non-critical task areas							26%					
Total gross area							100%					

* Areas can be m² or ft².

For each of the facade orientations, the Building Daylight Summary Table contains the three required sDA values, the two ASE values, the total glazing area, and the area-averaged blinds-operation fraction across the different facade orientations and their window groups. Blinds operation is separately reported for window groups that apply a 2% trigger versus an automated shading-control system. Window groups with diffuse glazing, to which no shading is applied, are included as a third category.

Software should provide at least eight different orientation groups for summarizing performance data across facades (preferably selected by the user). In addition, orientation groups can be added for skylight-illuminated areas and for non-daylit zones or interior zones that receive little or no daylight but are to be included in the total building sDA calculation. A conventional grouping of spaces would be the four orientations of a rectangular building. Corner offices with glazing on two different facades can be assigned to the orientation from which more daylight is expected to be received. For odd geometries, the orientations could be divided into 45-degree angular zones on a compass, or in the case of a building with a pentagonal shape, five facade orientations would be appropriate. ASE and sDA are tallied across the total analysis area assigned to each orientation, as well as for the entire building (all spaces that are included in the Space-by-Space Daylight Simulation Report).

For the daylight apertures across each orientation, the glazing area, blinds control type—2% trigger, automated, multiple, or none (where *none* applies only to diffuse glazing)—and the area- and time-averaged fraction of window area occluded by blinds are reported. Building totals for each column are then provided at the bottom of the table.

Glossary

A limited number of daylighting terms are defined in this glossary. Additional lighting and daylighting terms are defined in *ANSI/IES LS-1-22, Lighting Science: Nomenclature and Definitions for Illuminating Engineering*.¹⁰

analysis area: The horizontal work plane area for which a daylight calculation is performed. Calculation of this area is equal to the number of analysis points times the spacing distances for the rows and columns that were applied to create the analysis grid.

annual analysis period: The period considered for analysis. For a standard sDA_{300/50%} analysis, it is specified to be between the hours of 8 a.m. and 6 p.m. local clock time (10 hours per day, 365 days per year; see **Section 2.1**), with solar positions considered on the half hour using TMY weather data.

Annual Sunlight Exposure (ASE): The percentage of an analysis area that exceeds a specified direct-sunlight illuminance level more than a specified number of hours per year, with operable blinds left open (refer to **Section 3.0**).

automated daylight management system: Daylighting control devices whose position or light transmission level can be automatically changed by a control system to address sunlight penetration or perceived glare in a space. Acceptable automated daylight management systems include interior automated window blinds or shades; exterior automated louvers, shades, or blinds; or automatically controlled dynamic glazing. Automated methods of sunlight penetration control or glare control do *not* include manually operated interior or exterior shading systems; manually operated dynamic glazing; or fixed exterior overhangs, fins, shades, screens, awnings, or louvers whose position on the fenestration cannot be automatically changed or adjusted. Automated shading systems are permitted to have manual override but are required to default back to automated operation after a predefined period of no longer than two hours. Dynamic glazing is further defined by the National Fenestration Rating Council.¹¹

bidirectional scattering distribution function (BSDF): A matrix format describing the three-dimensional intensity of light distributed through a planar surface, typically used to describe the performance of a complex glazing system or light transmitting material. The file format was developed at Lawrence Berkeley National Laboratory (LBNL).

blinds: In this document, the word “blinds” refers to any operable daylight management device that can be applied to a daylight aperture. Examples include horizontal and vertical slatted blinds, roller shades, drapes.

Daylight Autonomy (DA): The percentage of time across a designated period in an annual analysis where the daylight illuminance exceeds a selected illuminance threshold value.

daylight factor: A measure of daylight illuminance at a point on a given plane, expressed as a ratio of the illuminance on the plane at that point to the simultaneous exterior illuminance on a horizontal plane from the whole of an unobstructed sky of assumed or known luminance distribution. Direct sunlight is excluded from both interior and exterior values of this illuminance.

daylight management device: Any device or glazing feature, whether interior, exterior, or embedded into the glass or glazing system, that can be adjusted to reduce or redirect daylight admitted to a space or to limit glare due to direct sunlight.

direct sunlight: An interior horizontal work plane measurement of 1,000 lux (100 fc) or more of direct-beam sunlight that accounts for window transmittance (including dirt depreciation), sky condition, and exterior obstructions (including fixed shading), but excludes the effect of any blinds, with no contribution from reflected light (i.e., applying a zero-bounce analysis), no contribution from sunlight that is scattered away from the direct beam, and no contribution from the diffuse-sky component.

illuminance threshold value: The illuminance level for Daylight Autonomy or Spatial Daylight Autonomy calculations, or for direct sunlight in an ASE simulation, that is used as a means of benchmarking a space’s performance.

occlusion area: The sum of the area of the sills, jambs, mullions, and any other opaque obstructions, measured in the plane of a window across the rough opening.

Perez All-Weather Sky Model: A method of describing the distribution of illuminance across the hemisphere of the sky for various weather conditions, based on five input parameters, first proposed in 1993.⁷

radiosity: A method of calculating the distribution of luminous flux in a space by solving a system of flux-balance equations within a space that has been subdivided into a set of surfaces, each having uniform exitance. In the standard version of this method, it is assumed that reflecting surfaces are perfectly diffusing, although there are extensions of the method for handling specular reflections and transparency (refer to *ANSI/IES LS-6-20, Lighting Science: Calculation of Light and Its Effects*¹² for further discussion).

ray tracing: A method of calculating the distribution of luminous flux in a space by determining the trajectories of a large number of individual “rays” of light. In *forward ray tracing*, rays originate at light sources, whereas in *backward ray tracing*, rays originate at a virtual viewpoint or sensor (refer to *ANSI/IES LS-6-20, Lighting Science: Calculation of Light and Its Effects*¹² for further discussion).

Spatial Daylight Autonomy (sDA): A measure of daylight illuminance sufficiency for a given area, reporting a percentage of floor area that exceeds a specified illuminance level (e.g., 300 or 150 lux [30 or 15 fc]) for a specified percentage of the analysis period (typically, 50% of the analysis time, which is 10 hours per day, 365 days per year; see **Section 2.1**). The standardized version of sDA requires blinds operation to limit direct sunlight penetration.

sunlight-tolerant area: A space, or an area within a space, that contains no fixed or critical visual tasks, and where the presence of direct sunlight is desirable. Often, these areas provide reflected daylight to neighboring critical task areas. These areas are excluded from both reported ASE and sDA simulations.

Typical Meteorological Year (TMY) data: Data that is generally collected by government entities and compiled into an hourly collection of sky and weather conditions that represent the general annual weather conditions at a site for use in building energy and daylighting simulations. Three versions are maintained

at the National Renewable Energy Laboratory (NREL), with slightly different formats and reference years. Other file types are available from other countries. Many software tools apply these data through standardized weather files, such as EPW files, that have been created for use in building energy simulations.

visible light transmittance (VLT): The percentage of the visible light spectrum (as measured in lumens) that passes through the center of a glazing system, for an incident direction normal to the plane of the window.

window group: A group of coplanar, or nearly coplanar, windows with similar shadow patterns from exterior shading and obstructions, with similar shading-device type and operation, that are associated with the same analysis area and will be controlled together. In generating a blinds-operation schedule, windows within ± 10 degrees of the same azimuthal orientation may be considered as part of the same window group. However, finer angular separation between groups is encouraged.

Annex A – History and Future

The development of sDA and ASE was driven by an initial intent to improve on the predictive performance of historical metrics such as daylight factor^{13,14,15} and single-point-in-time assessments of daylight adequacy, such as studying illuminance patterns only at noon on an equinox or solstice. The introduction of computer-based analysis by the 1980s added many opportunities and complications. The opportunity to fully consider the dynamics of climate and solar movement via the development of climate-based daylight analysis (CBDA)¹⁴ offered the potential to put daylighting analysis on an equal footing with dynamic analysis methods developed for hourly simulation of HVAC systems that are available in the marketplace.

However, as mentioned in the **Preface**, the sheer range and variety of inputs and analysis methodologies caused confusion in reporting and interpretation. Thus, the IES Daylight Metrics Committee was formed to help guide

the field of daylighting analytics toward an industry consensus of appropriate metrics and methodologies.

During the initial metric development process, existing daylight performance metrics were identified, reviewed, and assessed.² It was determined that no single metric could adequately address all of the dimensions involved in a successful daylighting system. The committee decided to first address the most basic and essential criteria—illuminance sufficiency and direct sunlight penetration—and then proceed to develop metrics for other dimensions as resources and research progressed. The committee also concluded that they should consider performance across whole architectural spaces (or a whole building), rather than individual occupant positions; that they should develop an annualized rating system customized to local climates rather than single-point-in-time assessment; and that they would need to account for the dynamic operation of window blinds.

In the metric development process, a variety of candidate climate-based metrics, including many variations of sDA and ASE, were compared to subjective evaluations of real spaces. Field research collected expert and occupant assessments of visual comfort in 61 daylight spaces across the USA and compared the results to detailed annual daylight simulation model outputs.² Since the original publication of IES LM-83-12, additional evaluations have been conducted on the initial data set and other original data sets.^{16,17,18,19,20,21,22,23} These results, combined with expert judgement accumulated through experience using these metrics, have informed the updates included in this edition.

There is not yet, however, a consensus on how to best predict the aggregated outcome of many individual decisions about the operation of blinds on windows as well as on skylights, although data on blind operation are available from numerous studies,²⁴ and multiple simulation algorithms for manual and automated blinds control have been proposed.¹⁵ In reality, variations in operation are likely to be a function of not only individual preferences but also device type, building type, occupancy patterns, building location, and design. Until there is a stronger research basis for predicting occupant interactions with window management systems, this document continues to take a greatly

simplified approach to blinds operation, considering only one factor that is common to most window management systems: the potential of direct sunlight to cause visual discomfort. Designers should understand that, in most cases, window management devices are a necessary component of a quality daylight delivery system to remedy problems that occur with direct sunlight penetration or excess daylight.

In the future, it is anticipated that experience using these metrics, along with ensuing published research, will inform refinements to the methodology and recommendations for their application. In addition, the committee anticipates subsequent metrics that will support evaluation of other dimensions of daylighting performance, such as uniformity, glare, circadian exposure, access to view, view quality, and supplementary electric lighting system operation.

The primary objective of these metrics is to enable an equitable comparison across spaces and buildings, using a continuous ranking rather than a pass/no-pass criterion. Thus, the committee has a continuing goal of defining a suite of metrics and supporting methodology that will enable professionals to discern nuances in the performance of daylit spaces and buildings along continuous scales in many dimensions.

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Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17th Floor, New York, NY 10005-4001. Email: standards@ies.org. Fax: 212-248-5017.

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